Armaan Sandhu

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Education

Worcester Polytechnic Institute | *Worcester, MA* **2025 - Present**

B.S. in Computer Science | GPA: 4.0 | Dean's List (Fall 2025, Spring 2026) | Expected May 2029

Carmel Junior College | *Jamshedpur, India* **2010 - 2025**

High School Diploma, CISCE Board | 97% (Unweighted) | Valedictorian, Class of 2025

Technical Skills

Languages: Python, C, C++, Rust, Zig, Java, x86-64 Assembly, JavaScript

Tools: Git, GDB, Linux, Bash, Valgrind, llama.cpp, Redis, PostgreSQL, SQLite

Concepts: OS Kernel Development, Systems & Concurrent Programming, Binary Analysis

AI/LLM: MCP, LLM Routing & Gateways, RAG, Vector Search, Tokenizers, Spec Driven Development

Open Source Contributions

AI Infrastructure (LiteLLM, LanceDB, Lance) **May 2026 - July 2026**

- Fixed a bug that silently dropped database rows during high-volume writes, a data-loss issue now patched in production for every user of the library
- Stopped a request-cancellation bug that was quietly locking teams out of their own API budgets
- Added support so Claude Code and other AI coding tools can connect through LiteLLM as a universal gateway

Systems & Browser Engines (systemd, CPython, Lightpanda) **May 2026 - July 2026**

- Fixed a reboot logic bug in systemd's update system, reviewed and merged by its creator, **Lennart Poettering**
- Built full file-upload support from scratch for an AI-agent browser engine, letting automated agents interact with file inputs like a real user would
- Landed a build-system fix into CPython, the reference implementation of Python itself

Over 20 pull requests merged across 6 major open-source repositories, spanning ***Rust, Python, Zig, and C.***

Experience

Founding Engineer & Product Head | *MLRouter* **June 2026 - July 2026**

- Built MLRouter, a single AI gateway that gives teams one endpoint to access 300+ models across every major provider, OpenAI-compatible so it drops into any existing stack with zero code changes
- Designed smart routing that picks the best model in real time by cost, latency, or capability, backed by automatic failover
- Built governance and security into the core: spend-capped API keys, full request-level observability, and compliance-ready controls (SOC 2, HIPAA-aligned)

Open Source Author & Maintainer | *mcp-persist* **May 2026 - Present**

- Built and maintain mcp-persist (**10k PyPI downloads in the first month**): production-grade MCP persistence layer across Redis, SQLite, and PostgreSQL with real-time streaming, PersistenceProxy, cross-backend migration, and a 500+ test suite; deployed in production by external teams.

Peer Learning Assistant | *WPI, Computer Science Department* **Incoming Aug 2026**

- Selected by CS faculty to tutor undergraduates in foundational computer science, data structures, and systems programming concepts.

Community Advisor | *WPI, Office of Academic Advising* **Incoming Aug 2026**

- Selected to support residential students through academic and personal challenges.

Research Experience

Independent Study, RunixOS | *WPI, advised by Prof. Craig Wills* **Early 2026 - July 2026**

- Built a capability-based operating system kernel from scratch as a formal Operating Systems research study, reworking how classic OS concepts like processes and memory protection work when nothing is trusted by default
- Engineered the kernel's core systems: a preemptible scheduler, synchronous and asynchronous IPC, and a capability-gated filesystem, all running behind an interactive console with built-in tracing and profiling tools

Projects

codebase-rag (Python) **July 2026 - Present**

- Built a tool that gives AI coding agents real understanding of a codebase, answering questions like "what calls this function" or "what breaks if I change it" in a single query instead of a pile of manual
- searches Automatically updates via git-hook, so the index never goes stale as code changes

rag-timetravel (Python) **June 2026 - Present**

- Built a debugger for AI search systems that lets you rewind any past query to the exact version of the data it ran against, then replay it to see exactly what changed and why results shifted over time
- Now the core engine that codebase-rag is built on

TokenDrift (Python) **June 2026 - Present**

- Built a safety check that catches breaking changes before they happen: measures how token counts, costs, and vocabulary shift when a team upgrades or swaps AI models
- Runs live inside MLRouter as a built-in safeguard before any model swap reaches production

Binary Exploitation (C, x86-64 Assembly) | *WPI, CS2011 Machine Organization & Assembly*

- Completed a full return-oriented-programming exploit chain against a live binary, including hidden stages, using GDB and objdump to reverse-engineer the target from scratch
- Went beyond the assignment and reverse-engineered the lab's own testing harness to understand how it validated exploits

Cache Simulator (C) | *WPI, CS2011 Machine Organization & Assembly*

- Built an N-way set-associative cache simulator with LRU eviction from scratch, then used its own output to drive real optimization decisions on a matrix transpose, verified with Valgrind traces

Awards & Recognition

- **Most Outstanding Student** | Carmel Junior College
- **International Silver Medalist & Bronze Medalist**, **National Gold Medalist (×2)** | Shotokan Karate India
- **Ni Dan (2nd Degree Black Belt)** | Shotokan Karate India - 10+ years competitive and instructing
- **Interschool Java Coding Competition** | 2nd Place (2023, 2024)